Using Generative AI to Strengthen Critical Thinking in High School

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GSCB 402 Innovative Design for Problem Solving

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Introduction

Generative AI is becoming a normal part of school and work. Students can use it to explain ideas, create examples, review writing, and practice new skills. These tools can make help available within seconds. Yet quick help does not always lead to learning. A student may submit better work while gaining fewer skills. High school teachers face a clear problem. They must prepare students to use an important tool without allowing that tool to replace the effort that builds sound judgment and critical thinking.

Critical thinking means more than finding an answer. It includes asking useful questions, weighing evidence, checking claims, finding bias, solving unfamiliar problems, and explaining why a choice makes sense. These skills grow when students must work through uncertainty and do more than just ask a computer for the answer. Generative AI can support that work by offering feedback or a different perspective, but it can also eliminate that work by providing a complete response before the student has formed an idea. AI has the potential to rob students of the choices that they make to better inform themselves.

How might I, as an educator, design high school learning activities that use generative AI to improve educational outcomes while strengthening rather than replacing students' critical thinking skills? The sources I found helped me understand the needs of students and teachers, and these steps helped define the problem as a lesson design issue rather than a simple choice between allowing and banning AI. The proposed classroom process is a prototype that teachers can test and improve.

The evidence supports a careful position. High school teachers should place generative AI inside a structured lesson process that protects independent effort, requires students to verify AI output, and grades the reasoning behind the final work. The research does not show that AI always helps or always harms thinking. It shows that guidance and task design matter. The sections that follow review the evidence, explain a teacher guided lesson protocol, and describe how a school can test it in a single unit before wider use.

Background and Literature Review

The broad research picture is mixed. Sun et al. (2026) reviewed 26 studies that reported 47 results about generative AI and critical thinking. The average result showed a moderate benefit. However, the range of likely results included both harm and benefit. The studies also differed in age group, subject, tool, and lesson design. The authors found early signs that the role assigned to AI and the rules for its use affected the results. Some groups contained too few studies for firm conclusions. This review warns teachers not to treat the average as a promise. AI can support critical thinking, but the conditions of use shape what students learn.

A large math study shows why classroom performance can be misleading. Bastani et al. (2025) studied almost 1,000 students in Grades 9 through 11. Students used standard resources, an OpenAI chatbot, or an AI tutor designed to provide hints. Both AI groups completed practice work more successfully than the control group. The open chatbot group improved its practice performance by 48 percent. Yet those students later scored 17 percent below the control group on an exam without AI. The guarded tutor further improved practice performance and largely prevented subsequent harm, but it did not yield a clear gain on the independent exam. Many students using the open chatbot asked for or copied full solutions. Better work during practice did not prove deeper learning.

The same risk appears in the prototype conversation. Eric Saxon, a quality assurance engineer, explained that software teams can produce more code with AI while reviewing less of the tool's output. That makes quality harder to protect because workers may not understand how the code works. He also argued that people need real experience with AI so they can learn when its output is useful and when it is weak. His main point was to keep human judgment involved and to check the system's outputs (E. Saxon, personal communication, September 1, 2026). This view cannot replace classroom research, but it does show that the habit of checking AI matters both after graduation and during school.

Research in high school English classes shows that students do not all use the same tool in the same way. Kwon et al. (2026) studied 55 Grade 10 students who used generative AI for resumes and practice interviews. Many students felt more confident and valued quick feedback. Some questioned the output, changed the wording, and made the work fit their own goals. Others accepted answers with little review or depended too much on the tool. Students also saw vague answers and raised concerns about honesty and copying. The study did not use a standard test of critical thinking and took place at a single school. It shows why access alone cannot produce a common learning result. Teachers must make active review part of the assignment.

Tang et al. (2026) found a similar pattern in one Grade 10 English class. Fifteen students used generative AI while studying poetry and a Shakespeare character. Strong questioning did not appear on its own. The teacher modeled how to ask questions, required students to check accuracy and bias, and gave them time to improve their prompts. Classmates also helped one another develop better questions. Subject knowledge mattered because students needed facts and context before they could challenge a weak AI answer. The small class and narrow setting limit the reach of the findings. Still, the study gives teachers useful actions. They can model

questioning, require proof, and ask students to explain how a later question improved the first answer.

Engeness (2025) adds evidence from 81 ninth grade students in six Norwegian classrooms. Three classes used an AI powered writing tool that identified ideas already covered and suggested ideas that might be missing. Three other classes used peer feedback without the AI tool. Both groups improved their essays. The AI group showed a stronger gain in writing quality and discussed essay content more often. Students in the comparison group focused more on spelling and word count. The AI feedback worked within peer discussion and teacher support, so the study does not show that automated feedback alone caused the gain. Agreement between graders was also moderate, limiting confidence in the magnitude of the difference. The useful point is that focused feedback can direct attention to ideas when people discuss and evaluate them together.

Generative AI may also help students manage their own learning. Ng et al. (2024) compared two chatbots during a three week science unit with 74 Secondary 4 students in Hong Kong. One chatbot used generative AI to guide goal setting, study plans, time use, help seeking, and self checks. The other chatbot followed fixed rules. Students using the generative tool gained more scientific knowledge and reported stronger motivation and greater control over their learning. The study covered two classes and a short period. Self regulated learning is also related to critical thinking, but it's not the same skill. The results still support a lesson design that asks students to plan, monitor their progress, and reflect rather than only asking for an answer.

AI literacy offers another part of the solution. Kong et al. (2024) studied a 14 hour project course with 128 senior secondary students. The course taught students to apply AI ideas to real problems, monitor their thinking, and consider ethical choices. Students improved their ability to

use AI ideas in problem solving. They also reported growth in awareness, confidence, and ethical understanding. The course focused on AI broadly, not only on generative AI. It did not show whether the gains lasted or moved into other classes. Its framework remains useful because it treats AI skill as more than just prompt writing. Students should understand the problem, explain their choices, and consider who may be helped or harmed by the result.

Together, the studies point to a common pattern. AI is most useful when it supports a learning process that remains visible to the teacher and student. It is risky when it hides effort behind a polished answer. The strongest warning comes from independent testing after AI is removed. The most useful positive studies include clear goals, guided feedback, discussion, checking, or reflection. No single study proves that one process will work in every subject or school. This gap supports a small classroom prototype. Teachers can apply the same core safeguards to different assignments, collect evidence from student work, and revise the process before using it more widely.

Solution and Discussion

The proposed solution is a five part teacher guided lesson protocol. It does not ban generative AI or allow open use for every task. Instead, it assigns the tool a limited purpose within a larger learning process. The protocol makes student thinking visible before, during, and after AI use. It also gives teachers evidence that is more useful than the final product alone. Each part addresses a risk or benefit identified in the research.

Independent Attempt

Students should begin with a short attempt before opening an AI tool. The attempt may be an outline, a solved example, a claim with evidence, a prediction, or a list of questions. It does not need to be complete. Its purpose is to activate prior knowledge and show what the student

can do alone. Teachers can keep this stage brief so it does not block useful support. A saved first attempt also gives the student something to compare later AI output against.

This step protects the effort that Bastani et al. (2025) found was easy to avoid when a chatbot supplied full solutions. It also gives students enough subject knowledge to question an answer, which Tang et al. (2026) identified as an important condition. If a student cannot begin the task, the teacher can provide a brief prompt or example before using AI. The goal is not to make students struggle without help. The goal is to prevent the tool from defining the problem and producing the first line of thought for them.

Defined AI Role

Before students use AI, the teacher should state what the tool may do. Useful roles include giving a hint, asking a question, offering feedback, suggesting an opposing view, or pointing out a missing topic. A role should match the learning goal. During a math lesson, AI might give the next hint without showing the full solution. During writing, it might identify an unclear claim without rewriting the paragraph. During social studies, it might offer a competing view that the student must test with sources.

A defined role keeps the student responsible for the main thinking. It follows the contrast between the open chatbot and guarded tutor in Bastani et al. (2025). It also matches the focused feedback used by Engeness (2025). Teachers should give students a sample prompt and model how to improve it. They should also state which uses are not allowed for that lesson. Clear limits make honest use easier because students do not have to guess where help ends and replacement begins.

Verification

Students should check every important AI claim before using it. A simple check asks four questions. Is the claim accurate? Does the evidence come from a trustworthy source? Does the answer fit the task and its context? Could missing information or bias change the result? Students should attach a source or brief note for each major fact they keep. They should correct or reject weak output rather than quietly remove it.

Verification turns AI output into material for judgment. It reflects the questioning process described by Tang et al. (2026) and the mixed student behavior found by Kwon et al. (2026). It also applies Saxon's workplace warning that an answer can be technically correct yet ineffective or unrelated to the real need. Teachers can model one verification in class and compare a strong check with a weak one. Students then see that fluent language is not proof.

Record Keeping

Students should keep a short record of their choices. The record can include the first attempt, the prompt used, one important AI response, the check performed, and the change made. Students should explain why they accepted, changed, or rejected the suggestion. This record should not include every message. It should capture the moments that shaped the final work.

The record gives teachers evidence of critical thinking that a finished answer may hide. It also helps students notice their own process. This supports the planning and reflection found in Ng et al. (2024) and the awareness of thinking studied by Kong et al. (2024). Grading the record gives students a reason to make careful choices. A student who finds and fixes a weak AI answer should receive credit for that judgment.

Independent Reflection and Assessment

The lesson should end with work that shows what the student can do without AI. The student might explain the main idea in class, solve a related problem, defend a choice, or answer a new question. A brief reflection should compare the first attempt with the final result. Students should name what they learned, what the AI did well, what it did poorly, and what they would do differently next time.

Teachers should grade both the product and the process. A practical score can include subject accuracy, strength of evidence, quality of verification, explanation of revisions, and independent understanding. The exact weights can change by subject. The key rule is that a polished product cannot earn full credit when the reasoning record is missing. This assessment addresses the gap between assisted performance and independent learning identified by Bastani et al. (2025).

Classroom Pilot

A teacher should test the protocol in one unit before using it across a course. The pilot can compare a short task completed before the unit with a similar task completed after it. Teachers should also review first attempts, reasoning records, verification notes, final work, and student reflections. These sources can show whether students are asking better questions and making stronger independent choices.

Student and teacher feedback should guide the next version. If the record takes too long, the teacher can reduce it to the most important choice. If students still request full answers, the allowed AI role may need stronger limits or better examples. If students verify only easy facts, the teacher can model how to test reasoning and bias. This cycle completes the design thinking

process. The classroom need is understood, the problem is defined, the protocol is created as a prototype, and evidence from the pilot is used to improve it.

The pilot should also test fairness and privacy. Every student needs a practical way to complete the assignment, even when access to an approved AI tool at home is limited. Teachers can provide class time, a shared school tool, or an equal option that does not require AI. Students should not enter private information into a public system. The teacher should record access problems and ask whether the tool worked equally well for different students. These checks connect the ethical awareness studied by Kong et al. (2024) with daily classroom choices. They also keep the pilot focused on student learning rather than on simple tool adoption.

Conclusion

Generative AI can support high school learning, but access alone does not build critical thinking. The research shows benefits when students receive focused feedback, plan their learning, discuss ideas, ask better questions, and reflect on their choices. It also shows a serious risk. Students can complete more rigorous practice work while learning less when AI supplies answers and reduces effort. Teachers need evidence of what students understand without AI.

The five part protocol offers a practical response. Students begin independently, use AI for a defined purpose, verify important output, record their reasoning, and finish with an independent check and reflection. Teachers grade both the process and the product. These steps keep human judgment at the center while still helping students gain useful experience with AI.

The next step is a small classroom pilot. One teacher can use the protocol in one unit, review student work, and compare independent performance before and after the lesson. Student feedback can show which directions are clear and which create confusion. The teacher can then revise the process and share the results with colleagues. This approach does not assume that one

rule will fit every subject. It gives educators a careful way to learn what works while protecting the thinking students will need in school, work, and daily life.

AI is the future. We must learn to evolve with technology, so it doesn't outgrow what we are capable of maintaining. The quicker we adapt and try to harness it as a tool, the better students and teachers will be in the future.

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